Transfer: waves and signals

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Choose appropriate wave models and explain what a signal establishes.

SUCCESS CRITERIA - I can...

1. Use wave speed and round-trip geometry correctly.
2. Separate frequency, amplitude and direction effects.
3. State the limits of an inference from a signal.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “frequency” and “wavelength” to explain your model or result. Sentence starters are optional.

frequency: number of cycles each second

wavelength: distance between corresponding points on repeated waves

round trip: travel from start to a reflector and back

01 Warm-up: bring back an earlier idea

Two targets each have range 500 m but different radial velocities.
Explain how they can have equal echo delays yet different Doppler shifts.

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $v = f \lambda$; $T = 1/f$; $d_{range} = v \Delta t/2$.
d_{range} is one-way range from round-trip time. Match speed and model to the wave and medium.

Worked reasoning: An echo time measures range using travel speed, while a frequency shift can indicate radial motion. Different observations answer different questions about the same target.

Your turn: Draw a measurement plan labeling outgoing pulse, return pulse and the time interval.

Sentence starter: My design shows … . It meets the requirement by … .



Marzano check - circle: ① ② ③

Transfer: waves and signals

antocciphysics

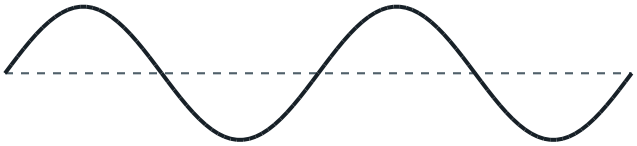
Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

Sketch a wave and label amplitude and wavelength 0.
Explain which tells about oscillation size and which about spatial repetition.

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .



Sketch and label your response.

Marzano check - circle: 1 2 3

04 Apply the physics

A radar receives an echo 0.012 s after transmission.
Use $c = 3 \times 10^8$ m/s to find range. The radar frequency is 3×10^9 Hz; find wavelength in vacuum.
Label the timing measurement and distinguish propagation speed from target motion.

Sentence starter: I found ... [unit]. This value means ... in this situation.



Marzano check - circle: 1 2 3

05 Explain your reasoning

A source gets louder while its frequency remains fixed. Does pitch necessarily rise?
Explain using wave properties.

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

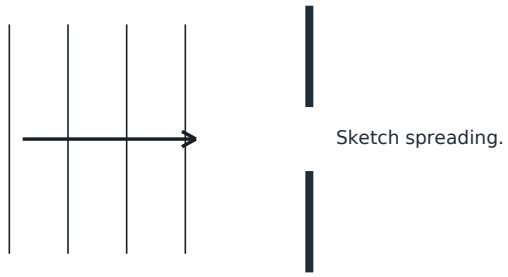
Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

At fixed wave speed, frequency becomes 4 times as large.
Compare wavelength and typical diffraction through the same opening.
Label wavelength and opening width; sketch the emerging wavefronts.

Word help: diffraction = wave spreading around an edge or through an opening

Sentence starter: Both ... . They differ in ... because ... .



Sketch spreading.

Marzano check - circle: 1 2 3

Transfer: waves and signals

antocciphysics

Continue down each column.

Sentence starters are optional. You may plan in any language.

07 Exit check

An ideal blackbody temperature doubles while a sound source frequency doubles. These are separate cases. **Compare** each wavelength change under its appropriate assumptions.

Word help: ideal = using stated simplifying conditions

Sentence starter: Both … . They differ in … because … .

Case	Quantity doubled	Wavelength comparison
Thermal spectrum	Temperature (K)	____
Sound, fixed speed	Frequency (Hz)	____

Marzano check - circle: 1 2 3

08 STRETCH: test the model

A stationary radar measures echoes at 10.0 and 9.6 microseconds separated by 0.2 s. **Find** both ranges and average radial velocity; distinguish this from the propagation speed.

Sentence starter: I found … [unit]. This value means … in this situation.



Marzano check - circle: 1 2 3